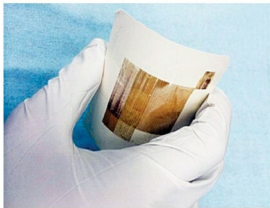
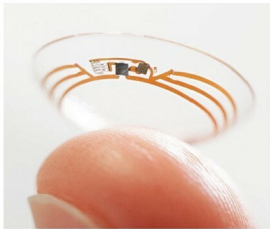


What is happening in India

With the aim to accelerate the development of flexible electronics industry in the country, Department of Electronics and Information Technology (DeitY), Ministry of Communication and Information Technology (MCIT), government of India, has established a centre of excellence for large-area flexible electronics (FlexE Centre) at IIT Kanpur. A dedicated team of 50 researchers and project staff will be working under the mentorship of IIT Kanpur's faculty members from different departments. Its current objectives include development of a national technology roadmap in association with academia, industry and public research organisations in the country, establishment of a broad research and development programme that serves as a foundation for development of domestic industry in this field.



High-mobility polysilicon layer was directly formed on paper by coating liquid silicon, which was annealed by pulsed laser-light by Prof. Ishihara and team at Delft University of Technology (Image courtesy: <http://optics.org>)



In association with pharmaceutical giant Novartis, Google is developing a smart contact lens to help patients manage diabetes. Apparently, Google has been granted a patent for the same (Image courtesy: www.forbes.com)

the form of measuring tears. With the help of miniaturised electronics, Google is testing a smart contact lens to measure glucose levels in tears using a tiny wireless chip and miniaturised glucose sensor embedded between two layers of soft contact lens material.

This novel contact lens will con-

tain a low-power microchip and a transparent ultra-thin and flexible electronic circuit, and will be used to measure blood-sugar levels of diabetes sufferers. Although these are still early days, Google is also planning to explore the possibility of integrating tiny LED lights to indicate to the user that glucose level has crossed or dropped below a certain threshold level.

Focus on hybrid and purely flexible systems.

Most recently, research has also shifted towards system integration, notes Petti. System integration is developing in both the directions of hybrid integration of rigid conventional silicon technology with flexible/printed electronics, as well as the realisation of fully-flexible and/or printable platforms.

The best examples of hybrid system integration can be surely found in the research done by Prof. Rogers at University of Illinois. Areas include flexible, stretchable, epidermal and biodegradable sensors and circuits using conventional silicon technology and unconventional substrates and architectures.

On the other side, a notable example of the fully-flexible approach includes the work by ETH Zürich (or Swiss Federal Institute of Technology Zurich) on lightweight and transparent metal-oxide electronics that can

wrap around a human hair. This can lead to fully-flexible and transparent smart contact lenses, as well as many other biomedical applications.

In the field of fully-flexible system integration, the biggest players are Interuniversity Microelectronics Centre (IMEC) and Holst Centre, who offer an extensive organic and metal-oxide based technology for flexible thin-film transistors, circuits, photovoltaics and active-matrix organic light emitting diode (AMOLED) displays.

Notable universities involved in research. Globally, key universities in this field are almost too numerous to name with each group working on topics ranging from fundamental science all the way to final product characterisation and prove-out, feels Farnsworth.

He says, "Other than California Polytechnic State University, some other groups doing important work in the area include CPI in the UK, Cetemmsa Technological Centre near Barcelona, Spain, VTT Technical Research Centre in Finland, Fraunhofer group in Germany, Industrial Technology Research Institute (ITRI) in Taiwan, The National Institute of Advanced Industrial Science and Technology (AIST) in Japan and EMSE near Marseilles, France."

He adds, "Dr Denis Cormier at Rochester Institute of Technology in Rochester, New York, is doing work combining printed electronics technologies with the still-developing additive manufacturing space."

Flexible electronics will make a ripple in every possible field

From healthcare monitoring, wearable and skin-like electronics, smart packaging, sensory tags for medical and biomedical applications, security and mobility commerce to environmental and industrial electronics, many novel applications are being envisioned. Petti says, "This is all enabled by the unique selling points of flexible and printable electronics, which include its mechanical flex-